

VINCENZO DI PERNA

Research Software Engineer | Distributed Systems, Blockchain & DLT

Location: Italy
Work authorization: EU citizen
Mobility: Open to relocation
Work mode: Remote / hybrid / on-site
Languages: Italian (Native) · English (Fluent)

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LinkedIn: [linkedin.com/in/vincenzo-di-perna](https://www.linkedin.com/in/vincenzo-di-perna)
GitHub: github.com/vindipe
Website/Portfolio: vindipe.github.io

SUMMARY

Research software engineer with a PhD in Blockchain & DLT and 4+ years of experience in distributed systems, automation, and data analysis. Developed Lilith, a topology-aware blockchain benchmarking framework that contributed to an ACM DEBS 2025 Best Student Paper and an ACM DLT 2026 journal article. International research experience across Italy, Switzerland, and the UK. Seeking engineering, R&D, and data roles in blockchain infrastructure and digital-asset analytics.

EXPERIENCE

- Digital Asset Researcher – DeFi & CBDC Research** 01/09/2024 - 31/10/2025
Cambridge Centre for Alternative Finance, UK
• Contributed to DeFi/CBDC research, launching the [Cambridge DeFi Navigator](#), validating on-chain datasets, and translating protocol analytics into reports for academic, institutional and policy stakeholders.
- Visiting Researcher – Blockchain Systems** 01/09/2023 - 01/06/2024
University of Neuchâtel, Switzerland
• Created Lilith, a topology-aware blockchain benchmarking framework, and ran large-scale experiments across public/permissioned blockchains, workloads and topologies.
- Doctoral Researcher – Distributed Systems & Blockchain** 01/11/2022 - 23/06/2026
University of Camerino, Italy
• Collaborated with an international team across systems research, experimentation, data analysis, technical writing, artifact release, and conference dissemination.
- Frontend Developer** 01/06/2022 - 01/09/2022
Reply
- Blockchain Development Intern** 01/10/2021 - 01/01/2022
SUPSI, Lugano, Switzerland
• Built a blockchain-based decentralised accounting and document-management prototype integrating Solid Pods and Web3 components for secure, user-controlled digital receipt management.

PROJECTS & DATASETS

- Lilith – A Topology-Aware Blockchain Benchmarking Framework** – [GitHub](#)
• Reproducible benchmarking framework to evaluate blockchain performance, energy consumption and variability.
- Cloud-Native Observability Lab** – [GitHub](#)
• Reproducible observability demo using Go, OpenTelemetry, Prometheus, Alertmanager, Loki, Tempo, Grafana, Redis, Docker Compose and Kubernetes.
- CryptoEntropy Explorer and Datasets for Economic Efficiency Analysis** – [GitHub](#) · [Explorer](#)
• Entropy-based dataset and explorer for analysing cryptocurrency efficiency across heterogeneous on-chain parameters.
- Blockchain Repeatability Dataset for Topology-Aware Benchmarking** – [GitHub](#)
• Large-scale repeated-run dataset capturing throughput, latency and energy variability across protocols, workloads and topologies.
- Decentralised Accounting & Digital Receipt Prototype** – [GitHub](#)
• Local demo with configuration validation, mock payment, PDF bill generation.

EDUCATION & CERTIFICATIONS

- Blockchain for Business (B4B)** 01/04/2023 - 01/06/2023
SUPSI, USI, FRANKLIN, Lugano, Switzerland
- Blockchain & Distributed Ledger Technology – Ph.D.** 01/11/2022 - 23/06/2026
University of Camerino, Italy
Thesis: Comparing Blockchains: Performance Energy, and Economic Efficiencies · [PDF](#)
- ICT Solutions Architect – M.Sc.** 01/10/2019 - 12/04/2022
University of Pisa, Italy
Thesis: Decentralized accounting system: Efficient document management system implemented on top of blockchain secure transactions environment · [PDF](#)
- Computer Systems and Network Security – B.Sc.** 01/10/2015 - 12/04/2019
University of Milan, Italy
Thesis: Threat Modeling Process: Strumenti per la gestione del processo di modellamento delle minacce · [PDF](#)

SELECTED PUBLICATIONS & RECOGNITIONS

- The Impact of Network Topology on Performance Metrics and Energy Consumption for Blockchains: Towards Repeatable Benchmarking – ACM-DLT Journal · [DOI](#)
- An Entropy-Based Approach to Evaluating the Economic Efficiency of Cryptocurrencies – DLT’25 · [DOI](#)
- Blockchain Energy Consumption: Unveiling the Impact of Network Topologies – IEEE ICBC’25 · [DOI](#)
- Impact of Network Topologies on Blockchain Performance (Best Student Paper Award) – ACM DEBS’25 · [DOI](#)
- Dynamic Taxonomy a Bridge from DeFi to TradFi – SSRN · [DOI](#)

CORE TECHNOLOGIES

- Programming:** Python, Bash; working knowledge of Rust, Go, TypeScript/JavaScript, Java, and C++
- Systems & Cloud:** Linux, Docker, Docker Swarm, Kubernetes, SSH, Git/GitHub, CI/CD, REST APIs, networking
- Data & Research:** Statistical Analysis, Experimental Design, Reproducible Research, Protocol Analytics
- Blockchain & DLT:** Ethereum, Solana, Algorand, Quorum, DeFi, CBDC, Smart Contracts, Tokenomics, DeFi
- Benchmarking:** Distributed Systems, Network Emulation, Performance Evaluation, Energy Measurement